GPU ALGORITHMS / RADIX EXPERIMENT

Stable GPU radix sort with recycled pages

99.6% less temporary storage, retaining 60.6% of Thrust's radix-sort throughput. The stable result returns to the original array.

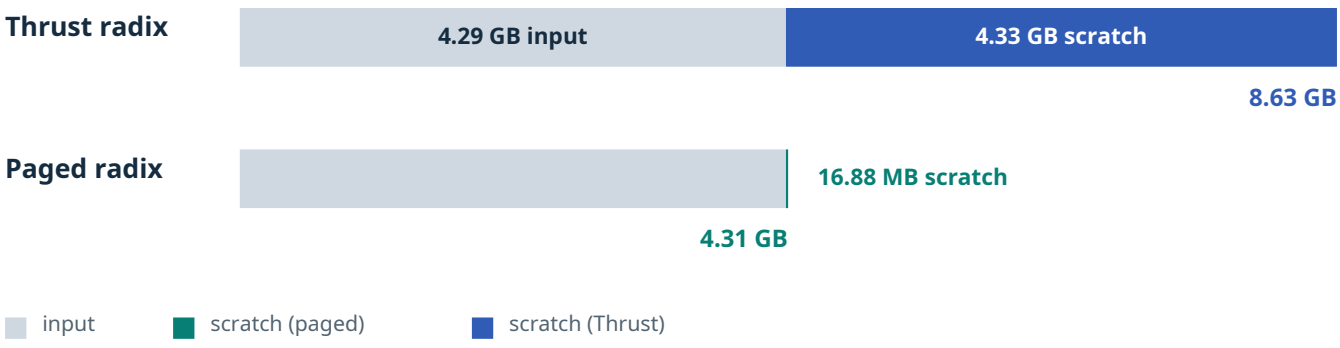
STABLE 32-BIT KEYS

14 SEPTEMBER 2026 • CCCL 3.6.0 CHECKOUT

16.88 MB	60.6%	4.31 GB
temporary storage	Thrust throughput retained	accounted peak memory

1,073,741,824 random unsigned 32-bit keys on one RTX PRO 6000 Blackwell. Thrust radix: **28.20 ms**; paged radix: **46.51 ms**. Stable ordering is checked separately using tagged records.

The memory difference



Peak = input + measured scratch. CUDA runtime/context and benchmark copies are excluded. Bars share a linear scale; the paged scratch allocation is the thin teal sliver.

CUDA 12.9 / 9 TRIAL MEDIANS / 5 WARMUPS / REUSED ALLOCATIONS

Final page permutation included; input resets and allocation growth excluded.